FUNCTIONAL ANALYSIS

(5 hours)

December 17, 2025

Please write on the front page your preferred time for the oral exam.

1. Let $X = l^1$ and

$$Y = \{x = (x_n) \in X : \lim_{n\to\infty} n^2 x_n \text{ exists and is finite}\}.$$

a) Show that a linear functional on Y can have at most one extension to a bounded linear functional on X

b) Let

$$l(f) = \lim_{n\to\infty} n^2 x_n, \quad x \in Y.$$

Does this functional have an extension to a bounded linear functional on X? Justify your answer.

2. Let $M : L^2([0,1]) \to C([0,1])$ be a bounded linear map.

a) Show that M has the form

$$Mf(x) = \int_0^1 K(x,y) f(y) dy, \quad f \in L^2([0,1]),$$

where $K(x,\cdot) \in L^2([0,1])$, for every $x \in [0,1]$, and

$$\sup_{x\in[0,1]} \|K(x,\cdot)\|_2 < \infty.$$

3. Let X be a Banach space and let $M : X \to X$ be a linear map. Prove that M is bounded if and only if there exists a set $S \subset X'$, dense in X', such that for each $l \in S$ the functional m_l defined by

$$m_l(x) = l(Mx), \quad x \in X$$

is continuous on X.

4. Let X, Y be Banach spaces and let $C : X \to Y$ be a compact linear map. Prove that if Z is a closed subspace of Y with $Z \subset R_C = C(X)$, then $\dim Z < \infty$.

Hint: Check if $C^{-1}Z$ is closed in X.

5. Let $T : l^2 \to l^2$ be defined by

$$T(x_n)_{n\geq 1} = (0, x_1, \frac{1}{2}x_2, \ldots, \frac{1}{n}x_n, \ldots).$$

Show that T is compact and determine its spectrum.

6. Let H be a Hilbert space and let $A, B \in \mathcal{L}(H)$ be injective compact symmetric linear maps such that:
(i) $\sigma(A) = \sigma(B)$,
(ii) All eigenspaces of A and B have dimension 2.
a) Prove that there exists a bijective isometry (unitary map) $U \in \mathcal{L}(H)$ such that $A = UBU^{-1}$.
b) Show by means of an example that the conclusion in a) may fail if we assume only the hypothesis (i).